

UniVerse

A Campus Companion

Profile

Viewing and editing the user's own account details.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

Owner: Ratul (notifications, profile)

Key files: features/profile/*

1. What it does

The Profile screen shows the logged-in user's details (name, role badge, academic/teaching info, avatar) and lets them edit allowed fields and sign out. It is a thin, focused feature - the textbook example of our layering.

file	job
<code>profile_service.dart</code>	The only Supabase layer: <code>fetchProfile(id)</code> and <code>updateProfile(id, changes)</code> . Returns the typed Profile model.
	Holds the Profile, loading/saving state; calls the service and notifies the screen.
<code>profile_screen.dart</code>	Pure UI: reads the controller, renders sections, triggers edit + sign-out.

2. The Profile model

`Profile.fromMap()` turns a `profiles` row into a typed object with helpers like `isTeacher`, `isStudent`, plus role-specific fields (`batch/section/student_id` for students; `teacher_code/department/designation` for teachers). The whole app reads the current user through this model - the `AuthController` exposes the raw map and features wrap it in `Profile.fromMap()`.

3. Update flow

- 1 User edits a field and saves.
- 2 `updateProfile(userId, changes)` runs an UPDATE on `profiles` WHERE `id = the user`, returning the fresh row.
- 3 RLS ensures a user can only update their OWN row.
- 4 The returned row is re-wrapped as a Profile and the screen rebuilds.

4. Likely defense questions

question	answer
Can a user edit someone	No - RLS scopes updates to <code>id = auth.uid()</code> ; the query also filters by the user's id.
Where does the role come	The <code>profiles.role</code> column, set at registration (or from the whitelist for admins).
Is this the same profile auth uses?	Yes - one profiles row per user, created at first login, read everywhere via <code>Profile.fromMap()</code> .